

# PALMER HUTT

Electrical Engineer | FAE | Solutions Engineer | Technical Sales

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## PROFESSIONAL SUMMARY

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Electrical Engineer with 7+ years across embedded firmware, hardware systems integration, and customer-facing technical delivery. Hands-on with ARM Cortex-M platforms, FreeRTOS, and industrial protocols including Modbus RTU, MQTT, CAN, and SPI. Proven track record in technical sales — translating complex system tradeoffs into client decisions and closed deals. The combination of embedded depth and commercial execution is the value: credible in a technical review and effective in front of a customer.

## TECHNICAL SKILLS

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**Firmware & Embedded:** C, C++, FreeRTOS (CMSIS-RTOS V2), ISR Design, Real-Time Systems, Static Memory, Hardware Watchdog (IWDG), Ring Buffer Design, ADC/DAC, GPIO, UART, I2C, SPI, CAN, PWM, DMA, Interrupt-Driven Architecture

**Hardware & Platforms:** ARM Cortex-M4, STM32 HAL, STM32CubeIDE, Oscilloscope, Logic Analyzer, Multimeter, Relay Control, CT Sensors, PZEM Power Modules, PCB Design, Schematic Capture

**Protocols & IoT:** Modbus RTU, MQTT, CAN Bus, BMS Architecture, REST APIs, JSON, Webhooks, Node-RED, Grafana

**Systems:** Solar & Battery Storage, EV Architecture, Battery Management Systems, CRM Integration

**Tools:** Git, MATLAB, VS Code, STM32CubeIDE

**Delivery:** Technical Sales, Systems Integration, Requirements Gathering, Remote Project Delivery, Stakeholder Communication, Cross-Functional Coordination, Technical Documentation

## PROFESSIONAL EXPERIENCE

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### Solutions Engineer | [EnergyPal](#) | May 2023 – Present

- Led end-to-end technical delivery of 100+ residential solar and battery storage deployments representing approximately \$4M in system revenue, averaging 8 kW solar with 14 kWh battery storage across SCE, PG&E, and SDG&E territories.
- Closed 15% of technical consultations by translating complex energy system tradeoffs — inverter sizing, grid interconnection, storage chemistry — into clear client recommendations and actionable deployment plans.
- Designed system architectures integrating solar arrays, hybrid inverters, and battery storage optimized for each site's energy profile and utility interconnection requirements.
- Implemented CRM-based project tracking and documentation workflows, improving forecasting accuracy and coordination visibility across distributed installation teams.

### Founder & Technical Lead | [The BuildTeam](#) | Mar 2020 – Apr 2023

- Founded and operated a remote embedded systems and custom electronics engineering practice — 40+ products delivered across 4+ product lines with 10,000+ customer reach and 99% satisfaction.
- Designed and delivered custom microcontroller-based systems including PWM-driven LED control architectures, sensor integration pipelines, and audio-reactive display firmware.
- Managed distributed client engagements across international suppliers and multi-timezone workflows with zero missed deadlines.
- Built technical documentation frameworks and hardware abstraction patterns enabling client-side maintenance without ongoing engineering support.

### Embedded & Systems Engineer | [Self-Employed](#) | Jun 2018 – Mar 2020

- Designed and delivered 10+ validated embedded builds across sensor, control, and display applications — each meeting defined functional and timing specifications.

- Resolved cross-domain hardware/firmware integration issues using structured iterative testing, logic analysis, and acceptance criteria.

## ENGINEERING PROJECTS

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### **Real-Time Energy Monitor — Modbus RTU + MQTT** — *STM32F401RE + FreeRTOS + PZEM-004T + ESP8266*

Production-style embedded IoT node reading six AC power measurements from a PZEM-004T energy meter over Modbus RTU — framing and CRC16 implemented from scratch, no library. Three-task FreeRTOS pipeline under mutex-protected shared data: Task A polls PZEM every 1s via USART1 with CRC validation and comms fault detection; Task B (high priority) monitors over-power, under-voltage, low power factor, and comms timeout thresholds at 100ms intervals, drives fault LED, kicks IWDG watchdog; Task C formats CSV telemetry to UART and JSON to ESP8266 for MQTT publish to HiveMQ. IWDG hardware watchdog tied to Task B resets system within ~2s on stall. Full engineering spec and demo at [huttllabs.com](http://huttllabs.com).

### **Deterministic Multi-Task Control Framework** — *STM32F401RE + FreeRTOS*

Real-time firmware on ARM Cortex-M4 using FreeRTOS CMSIS-RTOS V2. TIM2 ISR fires at 1 kHz, sampling ADC into a 64-sample static ring buffer with ISR execution under 50  $\mu$ s (<5% CPU budget). Three-task pipeline: Task A computes 16-sample rolling averages with stuck-sensor fault detection; Task B (high priority) drives relay and GPIO outputs within 100  $\mu$ s bounded latency; Task C streams UART telemetry every 500ms at 115200 baud. Zero dynamic allocation — all stacks, buffers, and queues statically allocated. Stack headroom stable at 130 words across full operating range. Fully verified on hardware. Demo at [huttllabs.com](http://huttllabs.com).

### **CAN Bus BMS Data Logger** — *STM32F446RE + FreeRTOS — In Progress*

CAN bus data acquisition system targeting EV battery management system integration. Hardware arrived: STM32F446RE, SN65HVD230 transceivers, OLED display. Phase 1: loopback validation. Architecture mirrors Modbus project — FreeRTOS task pipeline, mutex-protected shared struct, UART telemetry output.

### **Ford Ranger EV 2000 — Battery Modernization** — *Personal R&D*

Full battery system retrofit replacing legacy NiMH pack with modern lithium architecture. Scope includes chemistry evaluation, BMS selection and integration, pack sizing, thermal management, and compatibility validation against existing motor controller and vehicle electronics.

### **Capstone — Distributed Energy Storage in Active Distribution Systems** — *Ontario Tech University*

Modeled optimal sizing and siting of distributed battery storage to improve grid reliability and power quality across active distribution networks. Conducted standards research and integration analysis for distributed energy resources.

## EDUCATION & CREDENTIALS

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**BS Electrical & Electronics Engineering (Honours)** — Ontario Tech University, 2014–2018

**Engineer-In-Training (EIT)** — Ontario Society of Professional Engineers, Apr 2018